

EMG4 exploratory analyses

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Overview

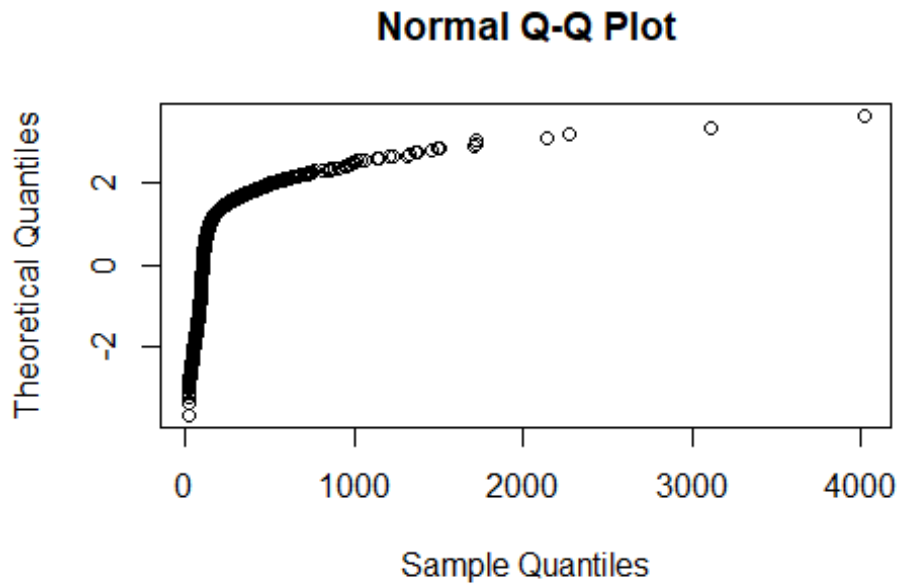
This document contains the exploratory statistical analyses of the EMG4 data combined with the embedded ratings, fairness and justice scores.

Load preprocessed MLM EMG data

1. Embedded Ratings

For exploratory reasons, we also look at the correlation between the EMG signal at the character morality segment and the ratings on the embedded rating task. We focus on the peak of the differential effect, which occurs 3-4s after the onset of the segment. The higher the rating score, the more moral the behavior was judged. So we expect to find a negative correlation, more frowning to immoral behavior (i.e. lower scores).

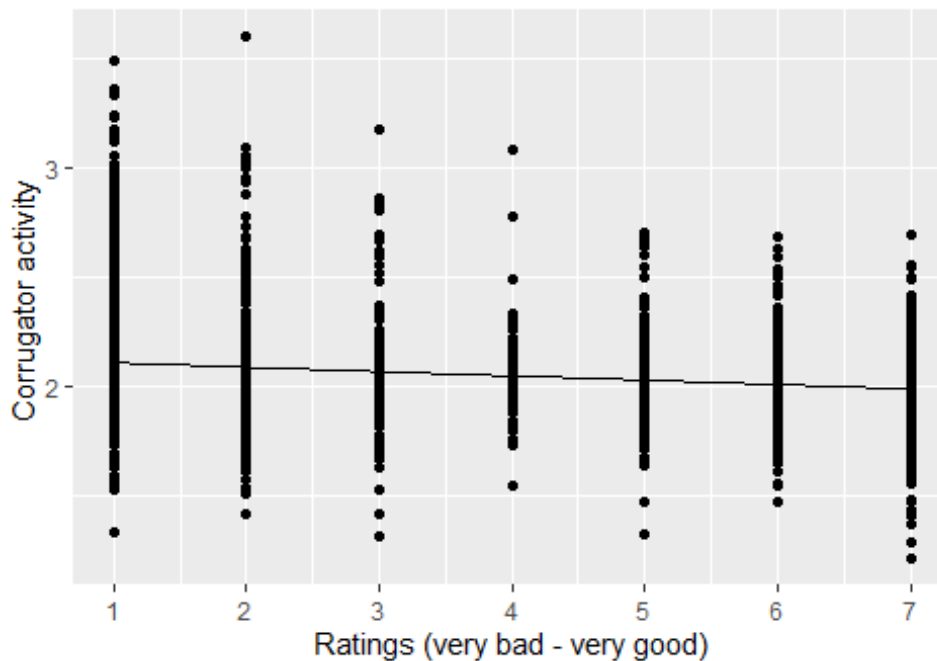
Check the normal distribution of the morality responses



Q-Q plot to check normality om EMG results at the Morality segment.

The averaged EMG data are not normally distributed, so we take the log transform.

Plot the relation between centered embedded rating and log corrugator EMG at the peak at the Morality Segment



Scatterplot between the centered embedded rating and log corrugator EMG at the peak at the Morality Segment.

Pearson's product-moment correlation: embedded ratings and morality EMG signal

```
##
## Pearson's product-moment correlation
```

```
##
## data: embed_rating and Morality_mean_log
## t = -14.175, df = 3699, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.2573231 -0.1962015
## sample estimates:
## cor
## -0.2269858
```

There is a weak negative correlation: $r(3699) = -0.2269858, p = 1.8672345 \times 10^{-44}$.

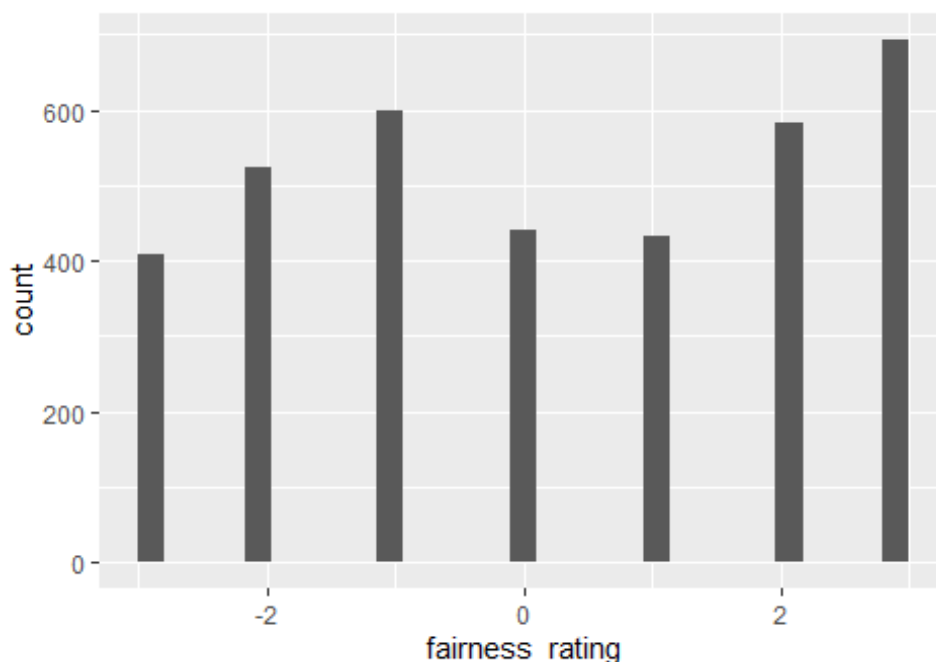
As expected, the lower the rating, the higher the corrugator EMG activation. There were some NA ratings and some trials were excluded in the baseline procedure. In total there was a data loss of 3.6197917%.

2. Fairness ratings

With the Story Fairness rating responses we were able to explore if and how the hypothesized effect of fairness-based evaluation would be reflected in a self-report measure as evaluation for this response tends to occur on a more conscious level. Therefore, we built a linear mixed model again focused on the interaction between the fixed factors morality condition (moral, immoral) and event valence (positive, negative) and with subject and item as random effects, but now with the Story Fairness ratings as the outcome measure. Finally, we performed pairwise comparisons to investigate significant effects.

A summary of the distribution and the means of the 4 conditions:

```
##
## -3 -2 -1 0 1 2 3
## 408 525 599 442 434 583 694
##
## -1.4375 -1.33224755700326 1.61195652173913 2.05901639344262
## 1 1 1 1
```



Histogram of Fairness ratings

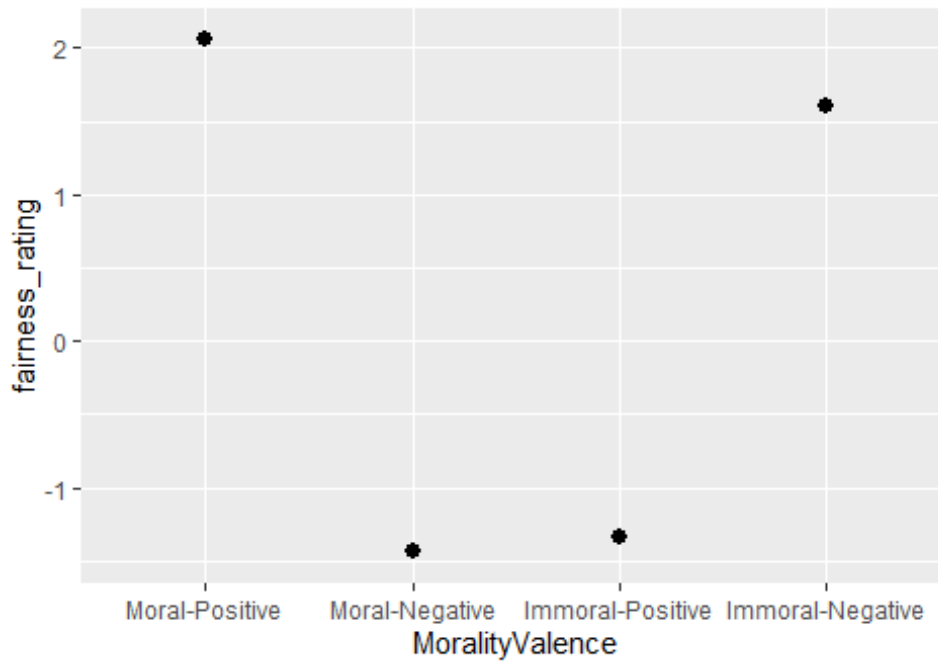


Figure on the mean ratings per condition

2.1 Building the Fairness rating model

Table 1 Fairness Ratings Linear Mixed Models Analysis Summary.

Predictors	Fairness Ratings (M1, <i>df</i> = 3677)		
	Estimates (95% CI)	<i>z</i>	<i>p</i>
(Intercept)	2.06 (1.97, 2.16)	42.42	<0.001*
MoralityCondition [Immoral]	-3.39 (-3.50, -3.28)	-60.08	<0.001*
StateAdjectiveCondition [Negative]	-3.50 (-3.61, -3.39)	-62.11	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	6.44 (6.29, 6.60)	80.85	<0.001*
Random Effects			
σ^2	1.46		
τ_{00} Markertrial	0.02		
τ_{00} sjCode	0.02		
ICC	0.03		
Deviance	11932.39		
log-Likelihood	-5966.19		

Note. Best model for the analysis of corrugator activity ($N_{\text{sjCode}} = 59$, $N_{\text{Markertrial}} = 64$). * significant *p*-value.

A combined amount of 87 ratings from 7 different participants were excluded from the analysis due to an error in the creation of the four versions of the Story Fairness Tasks (corresponding to the four versions of stimuli lists). These issues were quickly fixed so the data of all other participants was complete. Furthermore, there was 1 participant whose ratings on all 64 narratives were excluded from analysis as everything was rated as neutral. This resulted in a total data loss of 4.0625%.

Post-hoc analyses

```
## MoralityCondition StateAdjectiveCondition emmean SE df lower.CL upper.CL
## Moral Positive 2.06 0.0486 277 1.97 2.16
## Immoral Positive -1.33 0.0485 275 -1.43 -1.24
## Moral Negative -1.44 0.0483 273 -1.53 -1.34
## Immoral Negative 1.61 0.0485 276 1.52 1.71
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95

## StateAdjectiveCondition MoralityCondition contrast estimate SE
## Positive . Immoral - Moral -3.39 0.0564
## Negative . Immoral - Moral 3.05 0.0563
## . Moral Negative - Positive -3.50 0.0563
## . Immoral Negative - Positive 2.94 0.0563
## df lower.CL upper.CL t.ratio p.value
## 3572 -3.53 -3.25 -60.080 <.0001
## 3570 2.91 3.19 54.234 <.0001
## 3568 -3.64 -3.36 -62.113 <.0001
## 3567 2.80 3.08 52.238 <.0001
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
## Conf-level adjustment: sidak method for 4 estimates
## P value adjustment: sidak method for 4 tests
```

Plot of Fairness scores for the different State Adjectives for the Morality manipulations:

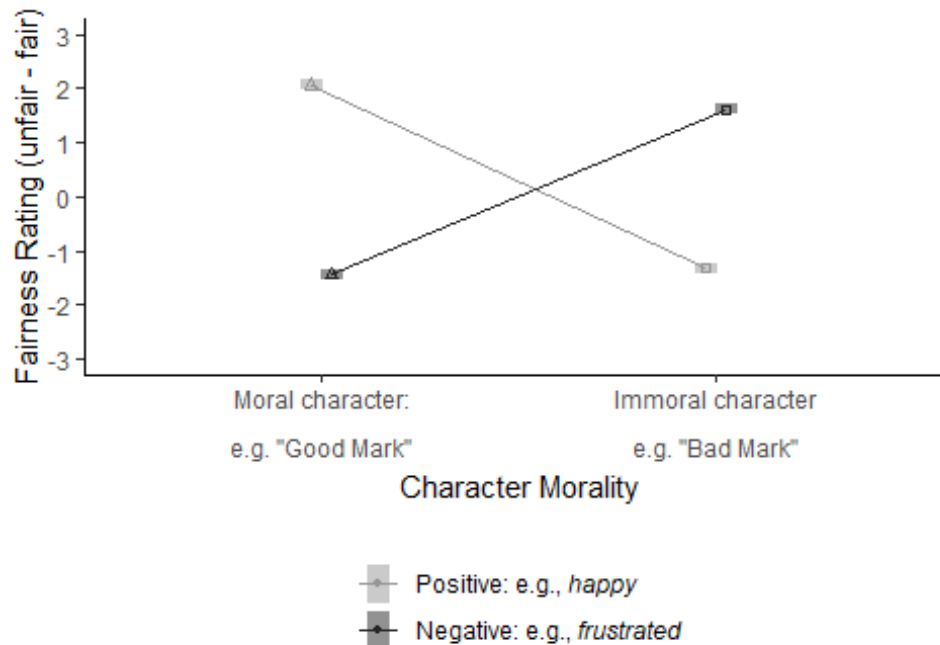


Figure on fairness scores for the different conditions.

3. Justice score

3.0 Load final MLM models from the regular analysis

3.1 Justice score and EMG signal at morality segment

The best model was Morality12: $\text{Morality_response} \sim \text{MoralityCondition} + \text{t1_immoral} + \text{t1_moral} + \text{t2_immoral} + \text{t3_immoral} + (1 + \text{MoralityCondition} \mid \text{sjCode}) + (1 \mid \text{Markertrial})$. We now add our justice score to that model and check whether it improves the fit.

```
load(paste0(getwd(), "/WORCS_EMG4/results/EMG4_MoralityJustice_final_model_60sjs.RData"))
load(paste0(getwd(), "/WORCS_EMG4/results/EMG4_Morality_js_model_60sjs.RData"))

anova(MoralityM12_js, MoralityM12_justice_score) # p = .09, so MoralityM12_js is the better model

## Data: justice_mor_data
## Models:
## MoralityM12_js: Morality_response ~ MoralityCondition + t1_immoral + t1_moral + t2_immoral +
t3_immoral + (1 + MoralityCondition | sjCode) + (1 | Markertrial)
## MoralityM12_justice_score: Morality_response ~ MoralityCondition * justice_score + t1_immoral +
t1_moral + t2_immoral + t3_immoral + (1 + MoralityCondition | sjCode) + (1 | Markertrial)
## npar AIC BIC logLik deviance Chisq Df
## MoralityM12_js 11 2384151 2384263 -1192065 2384129
## MoralityM12_justice_score 13 2384151 2384282 -1192062 2384125 4.7969 2
## Pr(>Chisq)
## MoralityM12_js
## MoralityM12_justice_score 0.09086 .
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.2 Justice score and EMG signal at state adjective segment

The best model is Adjective17: $\text{StateAdjective_response} \sim \text{MoralityCondition} + \text{StateAdjectiveCondition} + \text{MoralityCondition}:\text{StateAdjectiveCondition} + \text{t1_moral_neg} + \text{t1_moral_pos} + \text{t2_immoral_neg} + \text{t2_moral_neg} + (1 + \text{t1_moral_neg} + \text{t2_immoral_neg} \mid \text{sjCode}) + (1 \mid \text{Markertrial})$.

```
## Data: justice_adj_data
## Models:
## AdjectiveM17_js: StateAdjective_response ~ MoralityCondition * StateAdjectiveCondition +
t1_moral_neg + t1_moral_pos + t2_immoral_neg + t2_moral_neg + (1 + t1_moral_neg + t2_immoral_neg |
sjCode) + (1 | Markertrial)
## AdjectiveM17_justice_score_full: StateAdjective_response ~ MoralityCondition *
StateAdjectiveCondition * justice_score + t1_moral_neg + t1_moral_pos + t2_immoral_neg +
t2_moral_neg + (1 + t1_moral_neg + t2_immoral_neg | sjCode) + (1 | Markertrial)
## npar AIC BIC logLik deviance Chisq Df
## AdjectiveM17_js 16 665275 665418 -332621 665243
## AdjectiveM17_justice_score_full 20 665224 665402 -332592 665184 58.798 4
## Pr(>Chisq)
## AdjectiveM17_js
## AdjectiveM17_justice_score_full 5.187e-12 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Adding the justice score to the full factorial model improves the fit.

Table 2 State Adjective Segment Linear Mixed Models Justice Analysis Summary.

<i>Predictors</i>	State Adjective Segment (M17_full, df = 56080)		
	<i>Estimates</i> (95% CI)	<i>z</i>	<i>p</i>
(Intercept)	110.29 (94.11, 126.48)	13.36	<0.001*
MoralityCondition [Immoral]	1.13 (-2.19, 4.45)	0.67	0.51
StateAdjectiveCondition [Negative]	12.78 (9.07, 16.50)	16.75	<0.001*
Justice score	0.71 (-19.85, 21.27)	0.07	0.95
t1 moral negative	8.25 (-3.41, 19.92)	1.39	0.17
t1 moral positive	-5.47 (-8.94, -2.01)	-3.09	0.002*
t2 immoral negative	-11.65 (-35.02, 11.71)	-0.98	0.33
t2 moral negative	-34.89 (-43.89, -25.90)	-7.60	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	-0.93 (-6.53, 4.68)	-0.32	0.75
MoralityCondition [Immoral] * Justice score	9.47 (4.96, 13.98)	4.12	<0.001*
StateAdjectiveCondition [Negative] * Justice score	17.40 (12.92, 21.88)	27.61	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative] * Justice score	-18.40 (-25.60, -11.20)	-5.01	<0.001*
Random Effects			
σ^2	8150.90		
T00 Markertrial	112.97		
T00 sjCode	1793.42		
T11 sjCode.t1_moral_negative	1936.28		
T11 sjCode.t2_immoral_negative	7251.29		
ρ 01 sjCode.t1_moral_negative	0.17		
ρ 01 sjCode.t2_immoral_negative	-0.32		
ICC	0.20		
Deviance	665183.76		
log-Likelihood	-332591.88		

Note. Best model for the analysis of corrugator activity ($N_{sjCode} = 60$, $N_{Markertrial} = 64$). * significant p -value.

3.2.1 Probing the 3-way interaction

We can probe the 3-way interaction, provided we choose specific values for the justice scores. The scores range from -1.2352941 to 1.4705882, so we choose the values from -1 till 1.5 with increments of 0.5.

Table 3 Estimated Marginal Means for the State Adjective Segment at different levels of the Justice score.

StateAdjectiveCondition	MoralityCondition	justice_score	emmean	SE	df	lower.CL	upper.CL
Positive	Moral	-1.0	107.41	17.39	65.13	72.67	142.14
Negative	Moral	-1.0	102.79	17.40	65.27	68.03	137.54
Positive	Immoral	-1.0	99.06	17.40	65.17	64.32	133.80

StateAdjectiveCondition	MoralityCondition	justice_score	emmean	SE	df	lower.CL	upper.CL
Negative	Immoral	-1.0	111.92	17.08	60.47	77.76	146.07
Positive	Moral	-0.5	107.76	12.54	67.10	82.74	132.78
Negative	Moral	-0.5	111.84	12.55	67.40	86.80	136.89
Positive	Immoral	-0.5	104.15	12.54	67.14	79.13	129.18
Negative	Immoral	-0.5	116.51	12.34	63.48	91.85	141.16
Positive	Moral	0.0	108.11	8.17	71.01	91.83	124.40
Negative	Moral	0.0	120.90	8.19	71.78	104.57	137.23
Positive	Immoral	0.0	109.24	8.17	71.03	92.95	125.53
Negative	Immoral	0.0	121.10	8.10	70.17	104.94	137.25
Positive	Moral	0.5	108.47	5.60	69.42	97.30	119.64
Negative	Moral	0.5	129.95	5.63	71.03	118.72	141.18
Positive	Immoral	0.5	114.33	5.60	69.41	103.16	125.50
Negative	Immoral	0.5	125.69	5.63	71.07	114.46	136.91
Positive	Moral	1.0	108.82	7.14	71.88	94.59	123.06
Negative	Moral	1.0	139.01	7.17	72.87	124.73	153.29
Positive	Immoral	1.0	119.42	7.14	71.89	105.19	133.66
Negative	Immoral	1.0	130.27	7.11	72.10	116.11	144.44
Positive	Moral	1.5	109.18	11.21	68.04	86.81	131.55
Negative	Moral	1.5	148.06	11.22	68.39	125.67	170.46
Positive	Immoral	1.5	124.51	11.21	68.06	102.15	146.88
Negative	Immoral	1.5	134.86	11.05	64.89	112.80	156.93

Table 4 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Justice score.

contrast	justice_score	MoralityCondition	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	-1.0	Moral	-4.618	3.842	55918.53	-12.149	2.913	-1.202	0.229
Negative - Positive	-1.0	Immoral	12.857	4.671	11484.59	3.701	22.013	2.753	0.006
Negative - Positive	-0.5	Moral	4.082	2.805	55933.31	-1.417	9.581	1.455	0.146
Negative - Positive	-0.5	Immoral	12.355	3.342	13320.06	5.804	18.906	3.697	0.000
Negative - Positive	0.0	Moral	12.782	1.895	55934.15	9.068	16.496	6.745	0.000
Negative - Positive	0.0	Immoral	11.854	2.132	21814.14	7.675	16.032	5.560	0.000
Negative - Positive	0.5	Moral	21.482	1.388	55861.57	18.763	24.202	15.482	0.000
Negative - Positive	0.5	Immoral	11.352	1.393	55907.58	8.621	14.083	8.147	0.000
Negative - Positive	1.0	Moral	30.182	1.695	55935.87	26.860	33.504	17.807	0.000
Negative - Positive	1.0	Immoral	10.850	1.833	30817.48	7.258	14.443	5.919	0.000
Negative - Positive	1.5	Moral	38.882	2.536	55924.94	33.911	43.854	15.329	0.000

contrast	justice_score	MoralityCondition	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	1.5	Immoral	10.349	2.966	14789.07	4.534	16.163	3.489	0.000

3.2.2 Plotting the 3-way interaction

Plot of regression lines for the different categories of Morality and Valence.



Figure on State Adjective segment for different justice scores

People with high vs, low justice trait scores have different responses in the moral conditions than in the immoral conditions, as was indicated by the significant 3-way interaction (see Table 2, $b = -18.40$, $z = -5.01$, $p < 0.001$).

In the moral condition the participants who scored low on the justice trait score (i.e. people who do not agree to want positive things for good people and negative things for bad people) do not show a differential response towards the positive and negative state adjectives. Participants who were indifferent, or agreed to want positive things for good people and negative things for bad people, i.e. have a higher justice score, frowned more in response to negative state adjectives compared to positive state adjectives. As per our multiple-drivers model frowning at negative state adjectives for moral characters could be the net result of negative simulation and additional negative evaluation (unfair). Our justice trait score seems to influence the degree of frowning in this case, hinting at a possible attenuation of the negative evaluation. Only participants who value a 'sense of justice' frowned at unfair events that befall a moral character.

Interestingly, in the immoral conditions all participants, irrespective of their justice score, frowned more in response to negative state adjectives than to positive state adjectives. This pattern can be explained by negative simulation rather than evaluation, as the evaluation would counteract the simulation force yielding no differential responses or even a flipped response pattern. This is clearly not the case. However, the higher the justice score, the higher the overall response was (see Table 2, $b = 9.47$, $z = 4.12$, $p < 0.001$).

3.3 Justice score and EMG signal at affect reason segment

The best model is ReasonM22: AffectReason_response ~ MoralityCondition + StateAdjectiveCondition + MoralityCondition:StateAdjectiveCondition + t1_moral_neg + t1_moral_pos + t2_moral_neg + t3_moral_neg + (1 + t1_moral_neg + t3_moral_neg | sjCode) + (1 | Markertrial).

```
## Data: justice_reason_data
## Models:
## ReasonM22_js: AffectReason_response ~ MoralityCondition * StateAdjectiveCondition + t1_moral_neg + t1_moral_pos + t2_moral_neg + t3_moral_neg + (1 + t1_moral_neg + t3_moral_neg | sjCode) + (1 | Markertrial)
## ReasonM22_justice_score_alt: AffectReason_response ~ MoralityCondition * StateAdjectiveCondition * justice_score + t1_moral_neg + t1_moral_pos + t2_moral_neg + t3_moral_neg + (1 + t1_moral_neg | sjCode) + (1 | Markertrial)
## npar AIC BIC logLik deviance Chisq Df
## ReasonM22_js 16 1137482 1137634 -568725 1137450
## ReasonM22_justice_score_alt 17 1137319 1137480 -568643 1137285 165.04 1
## Pr(>Chisq)
## ReasonM22_js
## ReasonM22_justice_score_alt < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Adding the justice score to the full factorial model improves the fit.

Table 5 Affect Reason Segment Linear Mixed Models Justice Analysis Summary.

Predictors	Affect Reason Segment (M22_alt, df = 93483)		
	Estimates (95% CI)	z	p
(Intercept)	103.44 (88.34, 118.53)	13.43	<0.001*
MoralityCondition [Immoral]	3.21 (0.21, 6.22)	2.09	0.036*
StateAdjectiveCondition [Negative]	23.70 (20.34, 27.06)	13.82	<0.001*
Justice score	2.13 (-17.77, 22.03)	0.21	0.83
t1 moral negative	6.51 (1.26, 11.77)	2.43	0.015*
t1 moral positive	-3.82 (-5.70, -1.94)	-3.98	<0.001*
t2 moral negative	-14.56 (-17.47, -11.64)	-9.79	<0.001*
t3 moral negative	-4.72 (-9.35, -0.09)	-2.00	0.046*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	-13.29 (-17.83, -8.75)	-5.74	<0.001*
MoralityCondition [Immoral] * Justice score	18.29 (14.21, 22.37)	8.79	<0.001*
StateAdjectiveCondition [Negative] * Justice score	26.91 (22.86, 30.97)	13.01	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative] * Justice score	-38.46 (-44.28, -32.64)	-12.95	<0.001*
Random Effects			
σ^2	11154.41		
τ_{00} Markertrial	152.44		

T00 sjCode	1365.56
T11 sjCode.t1_moral_negative	83.73
ρ01 sjCode.t1_moral_negative	-0.01
ICC	0.12
Deviance	1137285.45
log-Likelihood	-568642.73

Note. Best model for the analysis of corrugator activity ($N_{\text{sjCode}} = 60$, $N_{\text{Markertrial}} = 64$). * significant p -value.

3.3.1 Probing the 3-way interaction

We can probe the 3-way interaction, provided we choose specific values for the justice scores.

Table 6 Estimated Marginal Means for the Affect Reason Segment at different levels of the Justice score.

StateAdjectiveCondition	MoralityCondition	justice_score	emmean	SE	df	lower.CL	upper.CL
Positive	Moral	-1.0	99.41	16.74	62.82	65.95	132.86
Negative	Moral	-1.0	96.19	16.75	62.94	62.72	129.66
Positive	Immoral	-1.0	84.33	16.74	62.86	50.86	117.79
Negative	Immoral	-1.0	106.29	16.75	62.90	72.82	139.75
Positive	Moral	-0.5	100.47	12.00	63.85	76.49	124.45
Negative	Moral	-0.5	110.71	12.01	64.11	86.72	134.71
Positive	Immoral	-0.5	94.54	12.00	63.89	70.56	118.52
Negative	Immoral	-0.5	110.72	12.01	63.94	86.74	134.71
Positive	Moral	0.0	101.54	7.70	66.98	86.16	116.91
Negative	Moral	0.0	125.24	7.72	67.64	109.83	140.65
Positive	Immoral	0.0	104.75	7.70	67.00	89.37	120.13
Negative	Immoral	0.0	115.16	7.71	67.05	99.78	130.54
Positive	Moral	0.5	102.60	5.12	74.15	92.41	112.80
Negative	Moral	0.5	139.76	5.15	75.83	129.51	150.01
Positive	Immoral	0.5	114.96	5.12	74.14	104.77	125.16
Negative	Immoral	0.5	119.60	5.12	74.15	109.41	129.80
Positive	Moral	1.0	103.67	6.68	68.83	90.35	116.99
Negative	Moral	1.0	154.28	6.70	69.71	140.92	167.64
Positive	Immoral	1.0	125.17	6.68	68.83	111.85	138.50
Negative	Immoral	1.0	124.04	6.68	68.79	110.72	137.36
Positive	Moral	1.5	104.74	10.70	64.43	83.36	126.11
Negative	Moral	1.5	168.81	10.71	64.73	147.41	190.21
Positive	Immoral	1.5	135.39	10.70	64.46	114.01	156.77
Negative	Immoral	1.5	128.48	10.70	64.43	107.10	149.86

Table 7 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Justice score.

contrast	justice_score	MoralityCondition	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	-1.0	Moral	-3.214	3.477	93371.46	-10.029	3.601	-0.924	0.355
Negative - Positive	-1.0	Immoral	21.960	3.432	93364.38	15.234	28.686	6.399	0.000
Negative - Positive	-0.5	Moral	10.243	2.539	93378.49	5.267	15.219	4.035	0.000

contrast	justice_score	MoralityCondition	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	-0.5	Immoral	16.186	2.447	93372.39	11.390	20.983	6.614	0.000
Negative - Positive	0.0	Moral	23.700	1.715	93375.34	20.339	27.061	13.821	0.000
Negative - Positive	0.0	Immoral	10.413	1.546	93380.51	7.383	13.442	6.737	0.000
Negative - Positive	0.5	Moral	37.157	1.255	93328.50	34.697	39.617	29.603	0.000
Negative - Positive	0.5	Immoral	4.639	0.989	93333.80	2.700	6.577	4.690	0.000
Negative - Positive	1.0	Moral	50.614	1.533	93376.56	47.609	53.619	33.011	0.000
Negative - Positive	1.0	Immoral	-1.135	1.325	93380.72	-3.733	1.463	-0.856	0.392
Negative - Positive	1.5	Moral	64.071	2.295	93374.51	59.573	68.569	27.919	0.000
Negative - Positive	1.5	Immoral	-6.909	2.173	93365.69	-11.169	-2.649	-3.179	0.001

3.3.2 Plotting the 3-way interaction

Plot of regression lines for the different categories of Morality and Valence.



Figure on Affect Reason segment for different justice scores

People with high vs, low justice trait scores have different responses in the moral conditions than in the immoral conditions, as was indicated by the significant 3-way interaction (see Table 5, $b = -38.46$, $z = -12.95$, $p < 0.001$).

In the moral condition, similar to the responses at the state adjective, the participants who scored lowest on the justice trait score do not show a differential response towards the positive and negative state adjectives. The other participants with a higher justice score, frowned more in response to negative state adjectives compared to positive state adjectives. As per our multiple-drivers model frowning at negative

state adjectives for moral characters could be the net result of negative simulation and additional negative evaluation (unfair). Our justice trait score seems to influence the degree of frowning in this case, hinting at a possible attenuation of the negative evaluation. Participants who stronger value a 'sense of justice' frown substantially more at unfair events that befall a moral character.

Interestingly, in the immoral conditions participants, with a low to moderate justice score, frowned more in response to negative state adjectives than to positive state adjectives. This pattern can be explained by negative simulation rather than evaluation, as the evaluation would counteract the simulation force yielding no differential responses or even a flipped response pattern. This is clearly not the case. However, for participants with a higher justice score the pattern starts to flip. For the participants with the highest justice score we see that they frown more in response to positive state adjectives than negative state adjectives (see Table 7, $b = -6.91$, $t = -3.18$, $p < 0.001$). This is the pattern that we had hoped to find overall and would indicate a counteracting force of fair evaluation and negative simulation of negative events befalling an immoral character.